

## Customer Churn Prediction Using Classical and Deep Learning Classification Models

Customer churn prediction is a central problem in subscription-based industries such as telecommunications, where retaining existing customers is significantly more cost-effective than acquiring new ones.

This project applies multiple classification techniques to a real-world telecommunications dataset and evaluates their relative performance, interpretability, and usefulness.

Dataset: Telco Customer Churn dataset from Kaggle, consisting of 7,043 customer observations with demographic, service, and billing variables.

Models Used: - Logistic Regression (GLM) - Random Forest - Support Vector Machine - Deep Neural Network

Evaluation: Models were compared using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Results: All models demonstrated meaningful predictive ability. Deep learning achieved the strongest performance, while logistic regression provided superior interpretability.

Conclusion: Customer churn can be effectively predicted using classification techniques. A balance between predictive accuracy and interpretability is recommended for real-world deployment.